THE NEW EUROPEAN SPACE AGENCY

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I. INTRODUCTION

On May 30, 1975, the representatives of eleven European governments[1] signed the new Convention for the establishment of the European Space Agency (ESA) in Paris.[2] This event was the culmination of a long period of deliberations to determine a European space policy and the instrument which will in the future execute this policy and the European space programs.

In 1962 two European Space Organizations were created: the European Space Research Organization (ESRO), responsible for satellite development; and the European Launcher Development Organization (ELDO), responsible for the development of European launchers. Whereas ESRO successfully developed seven satellites, launched about 150 sounding rockets and created its technical and operational establishments and stations, its sister Organization, ELDO, had to abandon its launcher development work in 1974.

The European governments participating in these Organizations were conscious of the need to redefine Europe's space policy and the European space programs in the science, application and launcher fields. To this end they established a European Space Conference in 1967. Meeting at Ministerial level, this Conference was charged with the elaboration of a new concept and the coordination of the work of the two Space Organizations until a single European space body could be formed. This purpose was achieved by the conclusion and signature of the ESA Convention. It should be noted, however, that this Convention has not yet entered into force as it is subject to ratification or acceptance by all Member States of ESRO and ELDO. The ELDO and ESRO Conventions will terminate on the date of the entry into force of the ESA Convention, and on this date the new Agency will take over all rights and obligations of ELDO and ESRO.

In order to ensure a smooth transition of functions from ELDO and ESRO to ESA, the ESRO and ELDO Member States decided that, whenever possible, the terms and conditions of the new ESA Convention should be applied, pending its entry into force.

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[1]These governments were: the Federal Republic of Germany, Belgium, Denmark, France, Ireland, Italy, the Netherlands, Spain, United Kingdom of Great Britain and Northern Ireland, Sweden, and the Swiss Confederation.

[2]Convention for the Establishment of a European Space Agency (signed Paris, France; May 30, 1975). For text of the convention, see 14 Int'l. Leg. Mat. 864 (1975).

The ESRO Member States decided, in addition, to change the name of that Organization to the "European Space Agency." Thus, work in the space field is executed now by ESRO which has conducted its activities under the name "ESA" since May 31, 1975, the legal basis being the ESRO Convention, but taking into account provisions of the new ESA Convention whenever legally possible.

II. THE STRUCTURE OF THE NEW ESA

The new ESA Convention reflects the classical structure of an intergovernmental organization: It will have a Council composed of delegates of the Member States; but a new idea has been introduced insofar as this Council can also meet at Ministerial level, thus taking over the role of the former European Space Conference. This Council is the legislative organ of the Agency. Its main tasks are: to define the policy to be followed by the Agency in respect of the accomplishment of its objectives; to approve the activities and programs of the Agency; to determine and review the level of resources to be made available by the Member States; to approve the annual work plans and budgets; to decide on the admission of new Member States; and generally to take all measures necessary for the fulfillment of the Agency's objective.

The Director General of the Agency is the executive organ; he is assisted by scientific, technical and administrative staff. He represents the Agency, takes all measures necessary for the management of the Agency, the execution of its programs and the implementation of its policy, in accordance with directives issued by the Council. The Director General has authority over the establishments of the Agency, and may submit proposals concerning activities and programs of the Agency to the Council. All functions of the Director General and his staff are exclusively international in character.

Both the Council and the Executive may create such advisory bodies as are necessary for the accomplishment of their tasks. Furthermore, the Council has to establish a Science Programme Committee, to which it shall refer any matter related to the mandatory scientific program and authorize it to make relevant decisions. The Council may also establish other subordinate Committees and delegate to them the power of decision; but such transfer of authority will have to be agreed by a two-thirds majority of all Member States. Today the main committees of this type are: the ESA Science Programme Committee; the Administrative and Finance Committee; the Industrial Policy Committee (responsible, *inter alia*, for taking decisions on the placing of industrial contracts); the International Relations Advisory Committee; the Advisory Group on Future Earth Resources Programmes; and the Advisory Group on the Use of Spacelab. In addition, the existence of particular Programme Boards should be mentioned. These are composed of delegates of participating States, which supervise the execution of the Application programs of the Agency, and which were established under a special legal regime on the basis of arrangements concluded among ESRO Member States participating in such programs and ESRO. They still continue their supervisory

activity and have in this respect authority to make decisions regarding these programs; for instance the adoption of the annual program budgets.

As far as the establishments of the Agency are concerned, at present they include the European Space Technology Center (ESTEC) located at Noordwijk, Netherlands; the European Space Operations Center (ESOC) located at Darmstadt, Germany; and the European Space Research Institute located at Frascati, Italy, which now functions as the Agency's Space Documentation Service. Furthermore the Agency has established several tracking and telemetry stations[3] and arranged for the use of national stations.[4] In addition, the Agency also possesses a launching base in French Guyana.

III. NEW FEATURES OF THE ESA CONVENTION

There is an essential difference between the ESRO and ELDO Conventions and the ESA Convention as far as mission and program aspects are concerned; the latter reflecting a new legal concept.

The purpose of the Agency is to provide for and promote, for peaceful purposes, cooperation among European States in space research and technology and their space applications, with a view to their use for scientific purposes and for operational space application systems. This "usage" and "application" effect is a new element which was not included in the former Conventions. To further this purpose, ESA's special task will be to elaborate and implement a long-term European space policy by recommending space objectives to the Member States, and by concerting the policies of the Member States with respect to other national and international organizations and institutions. This role of preparing space objectives and of coordinating the different interests of Member States in the international field is also new.

Furthermore, the Agency will elaborate and implement activities and programs in the space field, coordinate the European space program and national programs by integrating the latter progressively into the European program. Again this constitutes a new and important task. Also, the Agency will have to elaborate and implement an industrial policy appropriate to its program and recommend a coherent industrial policy to the Member States.

Among these mission aspects must be mentioned the Agency's role with regard to information and data dissemination. Indeed, the Agency must ensure that any scientific results be published and made widely available. Such technical inventions and data must be obtained from its contractors, as are appropriate for the protection of its interests and those of the States participating in the relevant programs, and of those of

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[3]These stations are located at Redu, Belgium; Odenwald, Germany; and Fairbanks, United States.

[4]The national stations are located at Cnes, France; and Fucino, Italy.

persons and bodies under their jurisdiction. The rights to be obtained for securing such interests shall include in particular the rights of access, of disclosure, and of use. Inventions and technical data will be communicated to the participating States. The Agency will furthermore disclose to Member States those inventions and technical data to which it holds proprietary rights, and they may be used for Member States' purposes or that of persons or bodies under their jurisdiction, free of charge.

A. Mandatory, Optional and Operational Activities

Turning to program aspects, one should first recall that the ESRO and ELDO Conventions were based on the concept of unity of program and membership. In other words, all Member States participated in the different programs. Only as an exceptional measure, based on a particular legal structure, was it possible to arrive at situations where some States did not participate in programs accepted by other States. This was the difficult legal situation which ESRO faced when it started its applications satellite projects,[5] the ARIANE launcher and the Spacelab development programs. Now, in the ESA Convention, account has been taken of the complexity of space activities and the different interests of States in space matters, by introducing the new concept of mandatory and optional activities. The mandatory activities include the basic activities, such as: education, documentation, studies of future projects and technological research work, the dissemination of information relevant to the harmonization of international and national programs, as well as the execution of the scientific program. All Member States have to participate in these mandatory activities, whereas in respect to optional activities they may formally declare themselves to be disinterested in participation. The ESA Convention mentions as optional activities: the design, development, launching and control of satellites and other space systems;[6] and the design, development and operation of launch facilities and space transport systems. When Member States agree to execute such optional activities they draw up a Declaration to this effect stating their undertaking, the financial conditions, the phasing of the program proposed, the scale of contributions, and furthermore establish implementing rules for the execution, by the Agency, of such a program. The Declaration and the rules are subsequently submitted to the Council for approval by a majority of all Member States.

A third category of activities is foreseen, called "operational activities". These are activities in the area of space applications which the Agency may carry out under conditions to be defined by the Council by a majority of all Member States. Examples of such activities are the placing at the disposal of operating agencies, such of ESA's own facilities as may be useful to them, and the launching and control of operational application satellites. These activities are not considered as Agency programs, and the cost of them is borne by the user concerned and therefore does not form part of the Agency's program budgets.

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[5] *E.g.*, aeronautical, telecommunications, and maritime satellite projects.

[6] *E.g.*, application satellites and systems.

B. Internationalization of National Programs

In relation to programs another interesting aspect should be underlined: The internationalization of national programs mentioned among the Agency's tasks will be implemented following a special procedure set out in an Annex to the Convention. According to this each Member State shall make available for participation by other Member States, within the framework of the Agency, any new civil space project which it intends to undertake, either alone or in collaboration with another Member State. With this end in view each Member State shall notify the Director General of the Agency of any such project before the beginning of its phase B (project definition phase), the timing and content of proposals for participation. It shall use its best endeavors to accommodate all reasonable responses, subject to agreement being reached, within the time-scale demanded by project decisions. The State shall subsequently submit a formal proposal to the Council when the project is to be executed as an optional program of the Agency. This special procedure underlines the importance Member States attach to the progressive integration into the Agency of national space activities.

C. Financial Planning and Budget

As far as the overall financial planning in respect of the Agency's mandatory and optional programs is concerned, the following should be noted: The Council shall determine, by a unanimous decision of all Member States, a level of resources for the coming five-year period, and shall, by the same majority, determine towards the end of the third year of each five-year period, and after a review of the situation, the level of resources to be made available to ESA for the new five-year period starting at the end of this third year. This procedure will certainly allow Member States and the Agency to establish an appropriately long term financial plan, with periodic review.

Each Member State contributes to the costs of the mandatory activities, and to the common costs of the Agency in accordance with a scale adopted by the Council by a two-thirds majority of all Member States. This scale will be based on the average national income of each Member State for the three previous years for which statistics are available. No State shall be required to pay contributions in excess of 25 percent of the total amount of contributions assessed by the Council to meet these costs. The cost of the optional activities are borne in the same manner, unless the States participating in a particular optional program decide otherwise.

From a budgetary point of view, one distinguishes now in ESA between a general budget (covering the above mentioned basic activities and fixed common costs, as well as non-fixed common costs and the support costs covering mandatory and optional programs), and program budgets for each program, whether mandatory or optional. The general budget and each program budget will be, under the final ESA regime, approved by the Council, by a two-thirds majority of the Member States and of the participating States respectively.

D. Miscellaneous Provisions

The main new features of ESA have been described above.In addition it should be noted that the new ESA Convention includes some other interesting provisions such as:

> (1) The Agency may cooperate with other international organizations and institutions and with non-member States, and conclude agreements with them to this effect. Such cooperations may take the form of participation by such States or organizations in one or more of the mandatory or optional programs mentioned before. Particular arrangements to give effect to such participation may provide that a non-member State shall have a vote in the Council when the latter examines matter pertaining to the progam in question.
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> (2) Cooperation with a non-member State may also lead to according it the status of "associate member" if that State at least undertakes to contribute to the financing of the studies of future projects. This underlines the openness of the new Agency towards international cooperation.
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> (3) The Agency will have international legal personality. Both ESA and its staff members, and the representatives of its member States, enjoy legal capacity, privileges and immunities in accordance with special provisions defined in an Annex to the Convention. Among these provisions the Agency's immunity from jurisdiction and execution, from requisition and sequestration of its property and assets, the exemption from national direct and indirect taxes, within the scope of its official activities, and from import and export duties, should be noted.
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> (4) A special International Arbitration Tribunal shall decide in case of disagreement between Member States or between them and the Agency, concerning the interpretation or application of the Convention. Any Member State of the Agency, whether party to the dispute or not, may intervene in the proceedings if the Tribunal considers that it has a substantial interest in the decision of the case. This Tribunal is also competant to resolve any conflicts among the Agency and Member States arising out of damage caused by the Agency, or involving any other non-contractual responsibility of the Agency, or involving the Director General or a staff member of the Agency in which the person concerned would claim immunity from jurisdiction. In respect to disputes arising between the Agency and the staff in respect of their conditions of service, provision is made for the establishment of an internal appeals board.

IV. CONCLUSION

The new European Space Agency must be considered, taking into account all the different political aspects which guided its creation, as a new instrument for the definition of a European space policy and as the principal means for the execution of programs decided in accordance with such a policy. It has, moreover, an important role to play in Europe's activities in the science and technology field, considering the impact of its program and budget. It will serve, also, as a lynchpin for the development of new projects and technologies. Finally, it will be used as a forum for the deliberation of international problems in the space field, in which its Member States are involved.

The new institutional framework described above[7] tries to combine different requirements: flexibility in the elaboration of new programs; efficiency in their execution; respect of different interests Member States have in the space field; acceptance of industrial return; sound equilibrium between legislative and executive tasks; and appropriate long term planning. It is hoped that the new Agency will, after its creation *de jure*, fulfill all these requirements in a satisfactory manner.

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[7]See Section II, *supra*.